

5	A	When the truck is empty, by Newton's second law, $F_{\text{net}} = m a$ $(-70\,000) = (5\,000) a$ $a = (-70\,000) / (5\,000) = -14.0 \text{ m s}^{-2}$ $t_1 = (v - u) / a = (30 - 50) / (-14.0) = 1.43 \text{ s.}$ When the truck is full, by Newton's second law, $F_{\text{net}} = m a$ $(-70\,000) = (5\,000 + 1\,300) a$ $a = (-70\,000) / (6\,300) = 11.1 \text{ m s}^{-2}$ $t_2 = (v - u) / a = (30 - 50) / (11.1) = 1.80 \text{ s.}$

		The difference $(t_2 - t_1) = 1.80 - 1.43 = 0.37$ s.
6	D	The equation is for the conservation of linear momentum in the vertical direction